



PROFESSIONAL CERTIFICATION

SAFE Structure Training

The SAFE Course covers foundation and floor system design software. Participants learn slab modeling reinforcement design and punching shear checks. Training includes post-tensioned systems soil-structure interaction and export to construction documents for building foundations.



28-35

Hours



Certified

Trainers



Industry

Expert



Professional Training | Industry-Recognized Certificate

Dubai, UAE

About Orbit Training Center

Your Gateway to Professional Excellence in Dubai



Who We Are

Established in 2023, Orbit Training Center is a leading professional training provider in Dubai, offering 60+ professional certification courses across diverse fields, including engineering, IT, design, and languages.



Our Benefits

-  Industry recognized certifications
-  Convenient location in the heart of Dubai
-  Affordable fees with multiple payment plans
-  Lifetime post-training support

★Our Features

-  Hands-on learning with real-world projects
-  Expert trainers with industry experience
-  Flexible learning options to fit your schedule
-  State-of-the-art infrastructure & facilities



Our Mission

We deliver outstanding courses, expert teaching, strong career guidance, and placement support to help every student achieve their best results. We stay connected with alumni through networking events, lifelong learning, and ongoing mentorship. We run our operations with smart systems and data-driven decisions to create a smooth, exceptional experience for all.

SAFE Structure Course

It is the tool to learn slab design with SAFE software.

Course Description

This course teaches SAFE for concrete floor and foundation design. Master finite element analysis deflection checks and reinforcement optimization. Projects cover mat foundations elevated slabs and PT systems for efficient structural engineering.

Duration

28-35

Hours of Training

Format

Classroom / Online

Flexible Option

Key Outcomes

- ✓ Master SAFE interface.
- ✓ Model floor systems.
- ✓ Analyze deflections loads.
- ✓ Design reinforcements.
- ✓ Check punching shear.
- ✓ Optimize PT tendons.
- ✓ Generate construction drawings.

Target Audience

- Structure Designers in slabs.
- Concrete Industry pros.
- Become Structure Engineer.

Certification

- Certificate of Course Completion
- Industry Recognized Certification
- Trained by Certified Professionals

Prerequisites

- Basic concrete knowledge needed.
- Computer with SAFE installed.
- Interest in foundation design.

SAFE Structure Course

Start with SAFE basics and slab modeling.

Intermediate

- Covers all Intermediate Levels
- Precise Geometry Creation
- Basic Parameters & Standards
- Understanding of Tools & Uses
- Intermediate Project Standard
- 1 Project at Completion

Duration
28 Hrs

Professional

- Covers all Professional Modules
- 16 Standard Professional Modules
- Learn with Industry Standard
- Professional Tools and Uses
- Multiple Professional Projects
- 3 Projects at Completion

Duration
32 Hrs

Advanced (Customized)

- Covers All Professional Modules
- 16+ Advance Professional Module
- Co-Ordination and Adv. Parameters
- Shop Drawings & As Built Drawings
- Govt. Standard and Submission
- 4 Projects in Different Categories

Duration
35 Hrs

SAFE Structure Course

Start with SAFE basics and slab modeling.

01 Introduction to SAFE Interface and Configuration

- > Customize toolbars ribbon shortcuts workspace
- > Configure units preferences precision colors
- > Master model explorer input editor tables
- > Set grid snap controls coordinate systems
- > Create foundation templates with standard settings

02 Importing References Defining Slabs Footings Geometry

- > Import scale DWG DXF PDF foundation plans
- > Define slab footing geometry boundaries openings
- > Establish levels grids reference planes
- > Model mat raft strip isolated foundations
- > Use generate replicate mirror commands efficiently

03 Material Properties Section Definitions Code Setup

- > Define concrete rebar soil subgrade materials
- > Specify code-based strength modulus coefficients
- > Create slab footing beam tendon sections
- > Assign thickness layers membrane bending properties
- > Select design codes ACI Eurocode BS IS

04 Modeling Slabs Footings Beams Walls Soil Supports

- > Draw edit slabs with thickness variations
- > Model footings beams walls parametrically
- > Assign soil springs area point line supports
- > Define subgrade modulus compression-only springs
- > Apply uplift checks nonlinear soil behavior

05 Load Definitions Patterns and Combinations

- > Define dead live soil pressure temperature loads
- > Create load patterns primary combination cases
- > Generate automatic seismic wind per code
- > Apply area line point distributed loads
- > Define service strength ultimate combinations

06 Advanced Analysis Setup and Options

- > Set linear nonlinear finite element analysis
- > Configure cracked section P-Delta soil interaction
- > Run static dynamic time history analysis
- > Specify mesh size auto-meshing refinement
- > Control iteration convergence solution parameters

SAFE Structure Course

Start with SAFE basics and slab modeling.

07 Analysis Setup and Options

- > Selecting analysis for foundations
- > Solver options for convergence
- > Soil springs and boundary conditions
- > P-Delta effects in foundations
- > Finite element methods integration

08 Static Analysis Techniques

- > Linear static for foundation equilibrium
- > Interpreting settlements and stresses
- > Bearing capacity checks
- > Uplift and overturning evaluation
- > Validating foundation assumptions errors

09 Dynamic Analysis Methods

- > Modal analysis for foundation frequencies
- > Response spectrum for seismic
- > Time history with motion inputs
- > Damping models in foundations
- > Scaling dynamic foundation responses

10 Design Codes and Parameters

- > Selecting ACI for foundation codes
- > Preferences for foundation materials
- > Overwrite options for elements
- > Auto-design iteration for foundations
- > Compliance checks and reports

11 Foundation Design Fundamentals

- > Isolated footing sizing and reinforcement
- > Strip footing design procedures
- > Mat foundation thickness determination
- > Punching shear checks
- > Settlement control measures

12 Advanced Foundation Design Processes

- > Raft foundation optimization
- > Pile cap and group design
- > Soil-structure interaction modeling
- > Seismic detailing for foundations
- > Reinforcement detailing and bars

SAFE Structure Course

Start with SAFE basics and slab modeling.

13 Results Interpretation and Output

- > Viewing foundation deformed shapes
- > Extracting soil pressures reactions
- > Generating foundation design reports
- > Contour plots for settlements
- > Exporting to CAD spreadsheets

14 Detailing and Reinforcement

- > Automatic reinforcement generation
- > Detailing drawings for foundations
- > Bar bending schedules creation
- > Reinforcement optimization
- > Detail export and documentation

15 Soil-Structure Interaction Modeling

- > Defining soil springs and subgrade modulus
- > Nonlinear soil behavior modeling
- > Pile foundation spring calculation
- > Interaction analysis and sensitivity
- > Advanced soil modeling techniques

16 Project Management and Collaboration

- > Worksharing and model coordination
- > Version control and model updates
- > Reporting and documentation workflows
- > Integration with BIM workflows
- > Professional best practices and case studies

Final Projects and Assessment:

- **Assessment:** Project-based evaluation through real-world projects to assess practical skills and application without theory exams.
- **Tailored Curriculum:** Customized to industry standards, covering essential topics for professional development in relevant fields.
- **Hands-on Projects:** Real-world simulations and interactive tasks to build applicable skills and create professional portfolios.
- **Certification Prep:** Targeted review sessions, practice exercises, and strategies for achieving industry certifications.
- **Ongoing Support:** Access to recorded materials, forums, instructor guidance, and career assistance for long-term growth.

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Muppuli Raja



One of great institute for REVIT BIM course. I completed my training with Mr Ahmed. He is one great trainer in this institute. He has very good knowledge for BIM Training. I completed my REVIT BIM training with orbit and it was very nice learning journey for me. And one of best thing is training center is very flexible for classes.



Arun



Best SolidWorks training in Dubai! The instructor made 3D modeling and assembly design easy to understand. The course covered all essential tools I need for product design and engineering work. Best training center for solidworks training. Best Training Center for Best Skills



Harshu CH



My experience at Orbit training center has been the best In every step I learn from there very clearly almost complete my course it's very interesting the trainer is the best professional and friendly ,and the rest of the team keep it up, thanks for supporting me Orbit and all staff...



Ranjan Axalan



I just finished my course in 3ds max. I learned a lot on how to create 3d model. Thank you for the experience and for the teaching method it help me to understand easily and grow my knowledge. I will use my learnings to my future company. Highly Recomanded!



Numan



I had good support from trainer Mr Ahmed and the other staff. They would schedule my BIM classes as per my convenience and would retake classes if I required. They always answer my calls when I need assistance. Not only that the trainer would go a mile a ahead by helping me with my work related issues like a true mentor.



Lelanie Fernandez



Overall, I highly recommend the Fusion 360 training at Orbit Center, especially with Ahmad as the instructor. This course has exceeded my expectations, and I am learning a lot. I would rate it 5 out of 5 stars!

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Aman Bakhtiyar



I had a great learning experience at Orbit Training Center. The trainers are knowledgeable, supportive, and always ready to help. The clear and practical teaching methods really helped improve my skills and confidence. I highly recommend Orbit Training Center to anyone looking to learn AutoCAD, Revit, and other professional skills.



Sana Zahid -



I recently completed the online Digital Marketing course with Orbit Training Center, and it was a great experience! The lessons were clear, practical, and easy to follow. The trainers were knowledgeable and supportive throughout the course. I've learned a lot that I can apply to my work here in Belgium. Highly recommend Orbit Training Center to anyone looking to improve their digital marketing skills!



Cleavon Henderson



Orbit Training Center offers outstanding CAD classes that are easy to understand, even as a first-year mechanical engineering student. The instructors simplify complex concepts and use practical examples, making the learning process engaging and effective. Whether you're new to CAD or improving your skills, Orbit Training Center provides a supportive and enriching environment. Highly recommended!



Danish Khan



Just finished Primavera P6 training at Orbit Training Center and it was fantastic! The trainer made complex scheduling concepts easy to understand with real-world examples. Got hands-on experience with project planning, resource allocation, and progress tracking. The course was well-structured and I feel confident using Primavera P6 for my projects now. Great value and would definitely recommend to other project managers!



500+ Students Trained

Join our growing community of professionals

95%

Satisfaction Rate

60+

Courses Offered

4.6

Average Rating



GET STARTED TODAY

Get in Touch with Us

At Orbit Training Center, we offer a diverse range of professional courses designed to equip you with the latest industry skills and knowledge. Our comprehensive programs cover in-demand fields such as Interior Design, Computer Science, Graphic Design, Digital Marketing, Solar Energy, and specialized software training including STAAD Pro, Revit Architecture, Tekla Structures, and more. Each course is crafted by industry experts to provide hands-on learning, real-world applications, and certification preparation.

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Free

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